Advanced Machine Learning, ETH Zurich

Hinge-chain networks

Any piecewise-linear function, one kink per layer

Carlos Cotrini

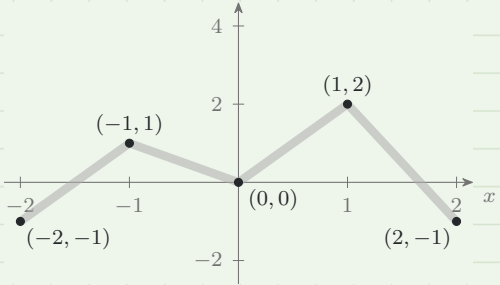
Autumn semester 2026

Entry 1

The goal

Our goal: a network whose output is exactly the zigzag

Our target is the zigzag f, which joins five points with straight lines on the interval from -2 to 2. We want a neural network that takes a number x and outputs $y = f(x)$, for every x in the interval.



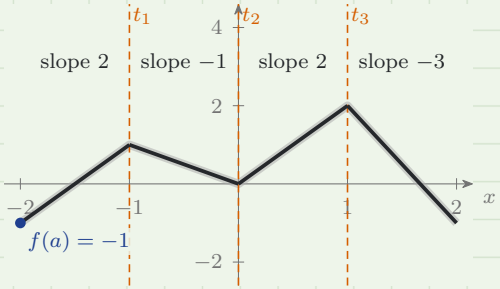
Grey: the target f, the zigzag.



In goes x, out comes y. We ask for $y = f(x)$ at every x from -2 to 2.

Exactly: an error of zero everywhere. We construct the network from ReLUs, piece by piece, with no training.

Start at the left end $a = -2$, at the height $f(a) = -1$, and walk to the right along a straight line. At a *kink* the slope changes, and the walk goes on straight again. The zigzag has three kinks, t_1, t_2 and t_3.



Grey: the zigzag. Graphite: the walk, one piece per click.

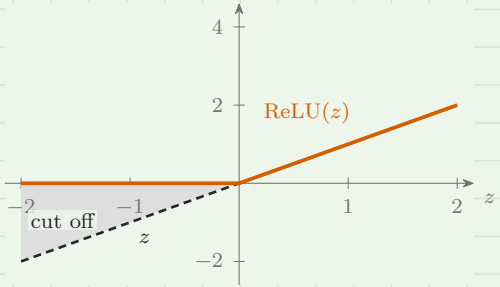
start value	$f(a) = -1$
slopes	$2,\ -1,\ 2,\ -3$
kinks	$t_1 = -1,\ t_2 = 0,\ t_3 = 1$

Such a function is *piecewise linear*: straight pieces that meet at kinks. Its start value, its slopes and its kinks pin it down. We build a network from exactly these numbers.

Entry 2

The hinge

The rectified linear unit, ReLU, is the activation of every neuron in our network. It passes a positive number on as it is and turns a negative number into 0: whatever falls below the floor at zero is cut off.



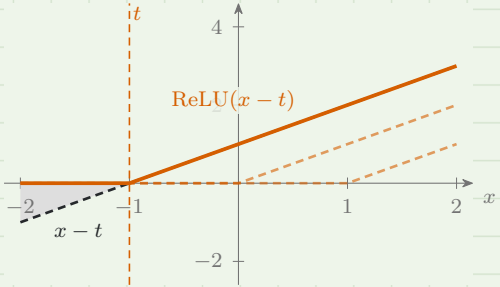
Dashed: the number z that comes in, plotted against itself.

Below zero, the floor cuts it off and puts 0 in its place.

Above zero, it passes on unchanged.

$$\mathrm{ReLU}(z) = \max\{z, 0\}$$

Feed the ReLU the input x minus a number t. Left of t, the number $x-t$ is negative and the floor sets it to 0; right of t, it rises with slope 1. We call $\mathrm{ReLU}(x-t)$ a *hinge*, and t its kink.

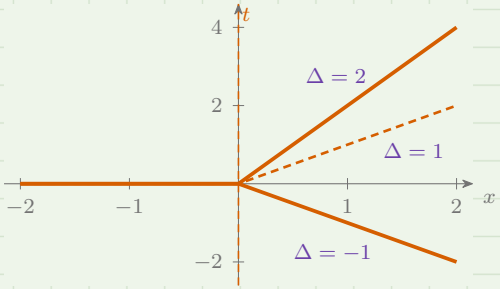


Dashed: the input shifted by t, the line $x-t$, here with $t=-1$.

The floor cuts what lies left of t. What is left is a hinge: flat at 0 up to its kink t, then rising with slope 1.

Changing t slides the hinge along x. Dashed: the same hinge at $t=0$ and at $t=1$.

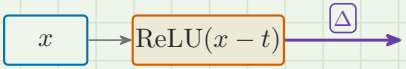
Multiply a hinge by a number Δ. Up to its kink it stays 0; after the kink its slope is Δ in place of 1. A positive Δ bends it up, a negative one bends it down.



Dashed: the plain hinge, $\Delta = 1$.

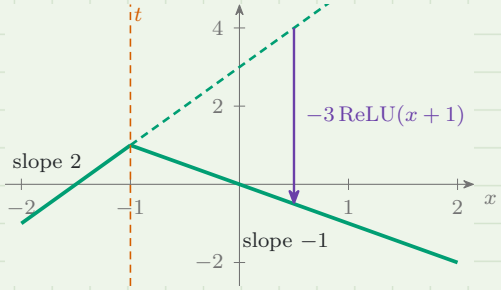
$\Delta = 2$: after the kink, the slope is 2.

$\Delta = -1$: after the kink, the slope is -1, and the hinge bends down.



In a network, Δ is the weight on the arrow that leaves the neuron, so it acts after the ReLU.

Add $\Delta\,\mathrm{ReLU}(x-t)$ to a line. Left of t the hinge is 0, so nothing changes; right of t the slope grows by Δ. Here a line of slope 2 gets the hinge $-3\,\mathrm{ReLU}(x+1)$.



A line of slope 2.

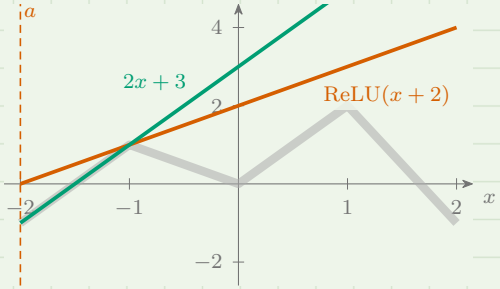
Add the hinge $-3\,\mathrm{ReLU}(x+1)$: its kink is $t=-1$ and its weight $\Delta=-3$. Left of -1 it adds nothing; right of -1 it pulls the line down (violet arrow).

Right of the kink the slope is $2+(-3)=-1$. The line now bends at t, and Δ is the change of slope there: a weighted hinge is a bend.

Entry 3

The hinge form

Put the kink of a hinge at the left end a. On $[a, b]$, with $b = 2$, the number $x - a$ is never negative, so the hinge $\mathrm{ReLU}(x - a) = x - a$ is a straight line. Times the first slope $\Delta_0 = 2$, plus $f(a) = -1$, it draws the first piece.

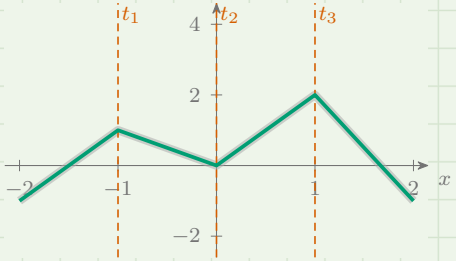


Grey: the zigzag. Vermillion: hinge 0. Green: the start value plus hinge 0.

Hinge 0 has its kink at the left end, $a = -2$. Right of a it never meets the floor: on $[a, b]$ it is the line $x + 2$, with slope 1.

Times the first slope $\Delta_0 = 2$, plus the start value -1: the line $2x + 3$. It is the first piece of the zigzag, continued straight on.

Start from $f(a) = -1$ and add hinge 0 for the first slope. Then add one hinge at each kink, weighted by the slope change there: the new slope minus the old one.



Grey: the zigzag. Green: the sum so far, one hinge more per click.

$$
\begin{aligned}
f(x) = {} & -1 && \text{start value } f(a) \\
& +2\ \mathrm{ReLU}(x+2) && \text{hinge 0: slope 0 to 2} \\
& -3\ \mathrm{ReLU}(x+1) && \text{kink } -1\text{: slope 2 to } -1 \\
& +3\ \mathrm{ReLU}(x) && \text{kink 0: slope } -1 \text{ to } 2 \\
& -5\ \mathrm{ReLU}(x-1) && \text{kink 1: slope 2 to } -3
\end{aligned}
$$

The violet numbers are the slope changes, $\Delta = (2, -3, 3, -5)$: the first slope, then the change at each kink.

Take any continuous piecewise-linear f on $[a, b]$ with k kinks $t_1 < \cdots < t_k$. It is its start value plus $k + 1$ hinges: hinge 0 at $t_0 = a$, which gives the first slope, and one hinge at each kink, which bends the curve there.

$$f(x) = \underbrace{f(a)}_{\text{start value}} + \underbrace{\Delta_0 \,\mathrm{ReLU}(x - t_0)}_{\text{hinge 0}} + \underbrace{\Delta_1 \,\mathrm{ReLU}(x - t_1) + \cdots + \Delta_k \,\mathrm{ReLU}(x - t_k)}_{\text{one hinge per kink}}$$

Δ_0 is the first slope, and Δ_ℓ is the slope change at the kink t_ℓ: the new slope minus the old one.

The zigzag: $k = 3$ kinks at -1, 0, 1; $f(a) = -1$; $\Delta = (2, -3, 3, -5)$.

Why it holds: both sides start at $f(a)$, and on the piece after kink t_ℓ both have the slope $\Delta_0 + \Delta_1 + \cdots + \Delta_\ell$.

Entry 4

One kink per layer

A layer with a single ReLU neuron turns the number u it receives into $\mathrm{ReLU}(wu + c)$. That map only rises (when $w \geq 0$) or only falls (when $w \leq 0$), and a chain of such maps still only rises or only falls.

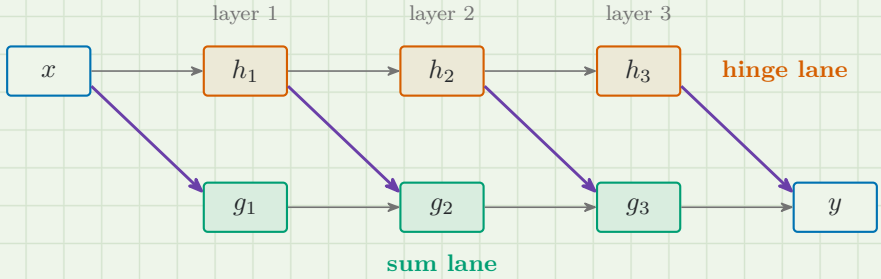


A rising map after a falling one falls; a falling map after a falling one rises. So the whole chain only rises or only falls, whatever its weights.

The zigzag rises, falls, rises and falls again: no chain of single neurons can draw it. We give each layer two neurons.

Two neurons per layer are enough, as the rest of the lecture shows. For functions of one variable, two is the least width that works (Hanin and Sellke, *Approximating continuous functions by ReLU nets of minimal width*, 2017).

Give each layer two ReLU neurons: on top a *hinge neuron* h_ℓ, which makes one kink, and below a *running-sum neuron* g_ℓ, which adds up the hinges handed down to it. Layer ℓ makes kink ℓ.

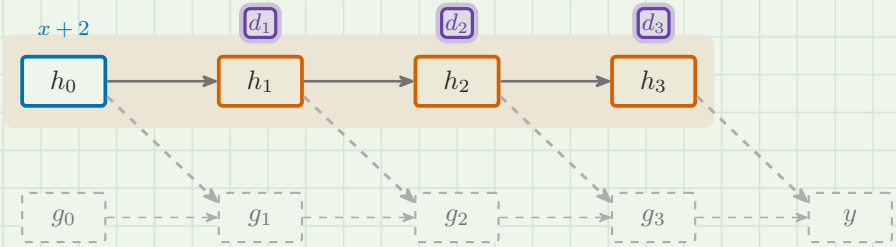


Hinge lane: each hinge neuron reads the hinge before it and makes the next kink.
Sum lane: each running sum passes its total on; the output y reads the last one.
Violet arrows: each hinge goes down to the next running sum, weighted by its slope change.

A chain of hinges above a chain of running sums: we call it a *hinge-chain network*.

Entry 5

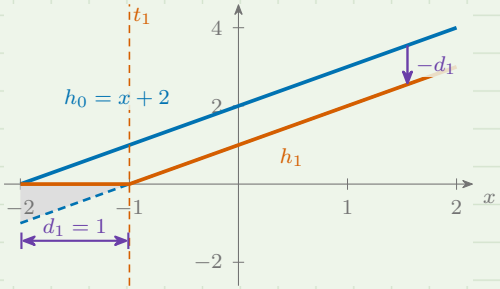
The hinge lane



So far. A piecewise-linear function is its start value plus one hinge per kink. The plan: a hinge lane that makes the kinks, and a sum lane that adds them up.

Now. The hinge lane. It starts from the ramp $h_0 = x + 2$, and each hinge neuron makes the next kink, one gap d_ℓ further to the right.

The hinge lane starts from the input shifted to 0 at the left end: the ramp $h_0 = x - a = x + 2$, which is hinge 0. The first hinge neuron lowers it by a *gap* $d_1 \geq 0$ and applies the floor: $h_1 = \mathrm{ReLU}(h_0 - d_1)$.



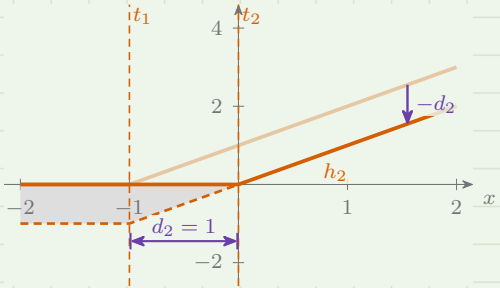
Blue: the ramp $h_0 = x + 2$, hinge 0, with its kink at the left end $a = -2$.

Lower it by the gap $d_1 = 1$: the line $x + 1$ (dashed).

The floor cuts off the part below zero (grey).

What is left is a hinge, $h_1 = \mathrm{ReLU}(x + 1)$, with its kink one gap to the right of a: at $t_1 = -1$.

The second hinge neuron lowers h_1 by the gap $d_2 = 1$ and applies the floor. The flat part sinks below zero and is cut back to 0: the hinge keeps its shape, and its kink moves right by d_2.



Start from the hinge h_1, kink at $t_1 = -1$.

Lower it by $d_2 = 1$: the flat part sinks to -1 (dashed).

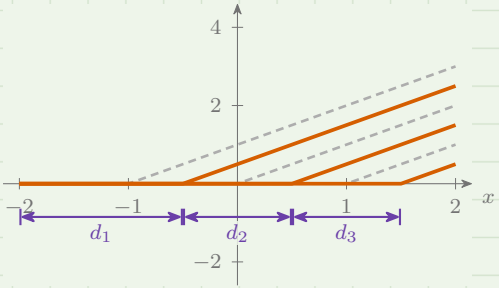
The floor cuts off all that is below zero.

Left: $h_2 = \mathrm{ReLU}(x)$, the same hinge, its kink moved right by d_2 to $t_2 = 0$.

$$h_\ell = \mathrm{ReLU}(h_{\ell-1} - d_\ell)$$

Every hinge neuron: kink $t_\ell = t_{\ell-1} + d_\ell$.

The kinks form a chain of steps from the left end a: kink ℓ sits ℓ gaps to the right of a. Make one gap longer, and every kink after it moves along.



Vermillion: the hinges h_1, h_2, h_3. Violet: the gaps.

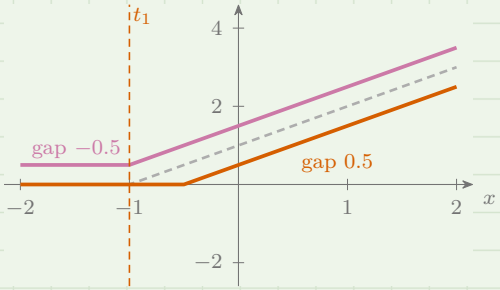
$$t_\ell = a + d_1 + d_2 + \cdots + d_\ell$$

The zigzag: $d = (1, 1, 1)$, so the kinks sit at -1, 0 and 1.

Make the first gap longer, $d_1 = 1.5$: all three kinks move right by 0.5. The grey hinges are where they were.

A gap is never negative, so the kinks stay in order. On the page, dragging kink ℓ in the plot sets its gap d_ℓ.

A gap lowers the hinge before it, and the floor cuts the sunken flat part back to 0. A negative gap would raise the hinge instead: its flat part stays above zero, the floor cuts nothing, and what comes out is no hinge at all.



Start from the hinge h_1, kink at -1.

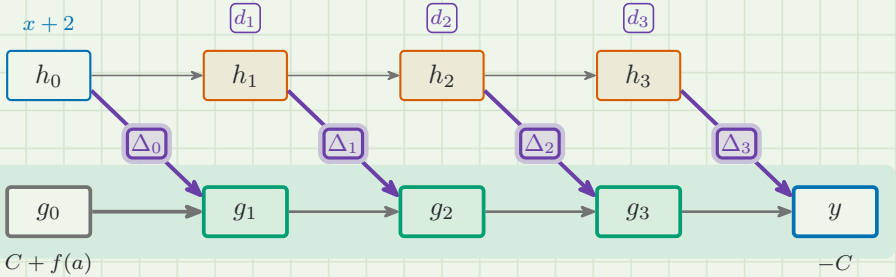
A gap of 0.5: a hinge again, its kink moved right to -0.5.

A gap of -0.5 raises the hinge by 0.5. The floor cuts nothing, and the flat part stays at 0.5. A hinge is 0 left of its kink, so this is no hinge.

So every gap is at least 0, and the kinks come in order. The page keeps each gap between 0 and 4.

Entry 6

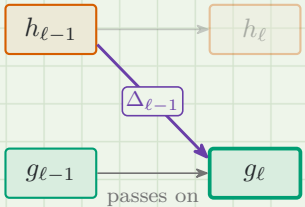
The sum lane



So far. The hinge lane makes one hinge per layer, $h_\ell = \mathrm{ReLU}(x - t_\ell)$, each kink one gap d_ℓ after the one before.

Now. The sum lane. Its running sums add the hinges up, each weighted by its slope change Δ_ℓ; a lift C keeps them above the floor; the output y adds the last hinge.

The sum lane keeps a running total of the hinge form, starting from the start value, $g_0 = f(a)$. Running sum g_ℓ passes the total $g_{\ell-1}$ on and adds the hinge before it, $h_{\ell-1}$, weighted by its slope change $\Delta_{\ell-1}$.



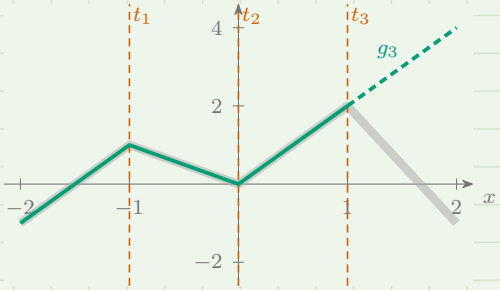
Grey arrow: the total passes on with weight 1.
Violet arrow: the hinge comes in, weighted by its slope change.

$$g_\ell = \mathrm{ReLU}(g_{\ell-1} + \Delta_{\ell-1}\, h_{\ell-1})$$

For now, suppose that no total dips below zero, so the floor does nothing.

The first one, for the zigzag:
$g_1 = -1 + 2\,(x+2) = 2x + 3$, its first piece.

Running sum g_ℓ holds the start value and the hinges 0 to $\ell - 1$. So it follows the zigzag up to the kink t_ℓ, where the next hinge would bend it, and goes on straight from there.



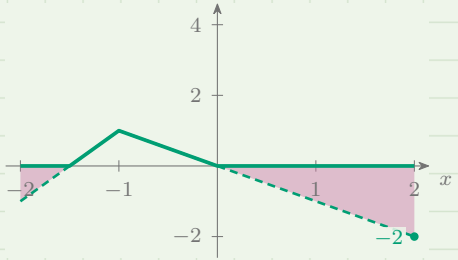
Grey: the zigzag. Green: one running sum per click, dashed where it leaves the zigzag.

g_1: the start value and hinge 0. It is the zigzag up to $t_1 = -1$; then the line $2x + 3$ goes straight on.

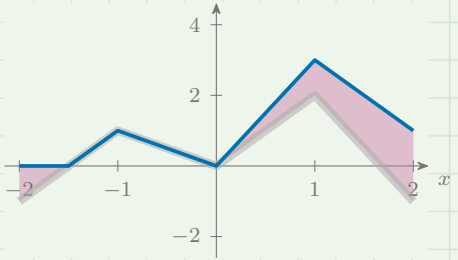
g_2 adds hinge 1: the zigzag up to $t_2 = 0$, then straight on with the slope -1.

g_3 adds hinge 2: the zigzag up to $t_3 = 1$, then straight on with the slope 2. The last hinge comes in at the output.

The running sums are ReLU neurons too, so each has a floor at zero. The zigzag's totals dip below it: g_2 should hold -1 at the left end and -2 at the right end. The floor cuts those parts off, and nothing later brings them back.

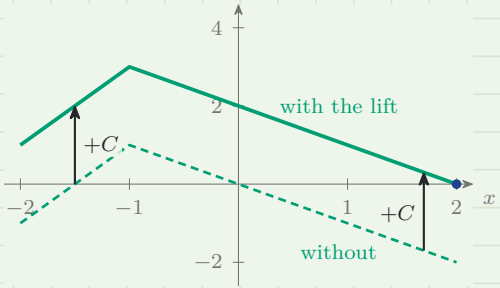


Dashed: the total g_2 should hold. Solid: what passes the floor. Pink: cut off and lost.



With no lift, the output y misses the zigzag (grey) wherever a total was cut: pink, the error.

Start the sum lane with a deposit, a *lift* $C \geq 0$. Every total is then C higher and stays above the floor; the output pays C back. The deepest dip is -2, so the zigzag needs $C = 2$.



Green: the running sum g_2, dashed without the lift, solid with $C = 2$.

Without a lift, g_2 would dip to -2 at $x = 2$.

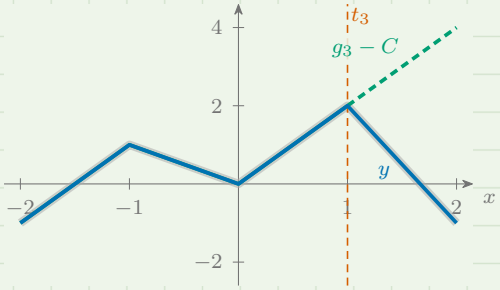
$$g_0 = C + f(a)$$

With $C = 2$, g_2 stays at or above zero, and so do g_1 and g_3.

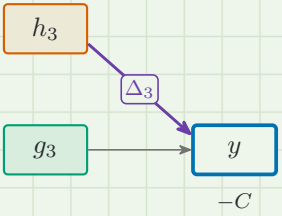
$C = 2$ is the least lift: g_2 just touches the floor at $x = 2$.

Moved right by 2 and up by 3, the same zigzag lives on $[0, 4]$ above the floor: no total dips there, and $C = 0$.

After the last layer, g_3 holds the zigzag up to the last kink, lifted by C. The output adds the last hinge, weighted by $\Delta_3 = -5$, and takes C off. It has no floor: y is a plain sum.



Green: $g_3 - C$. Blue: the output y, on top of the zigzag.



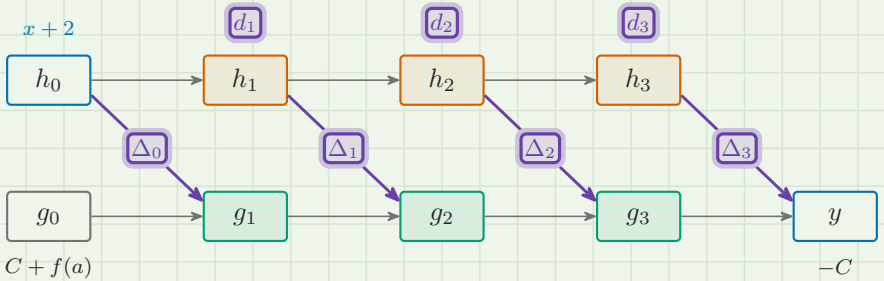
$$y = \Delta_k \, h_k + g_k - C$$

At $t_3 = 1$ the slope turns from 2 to $2 + (-5) = -3$.

So $y = f$: the network draws the zigzag exactly.

Entry 7

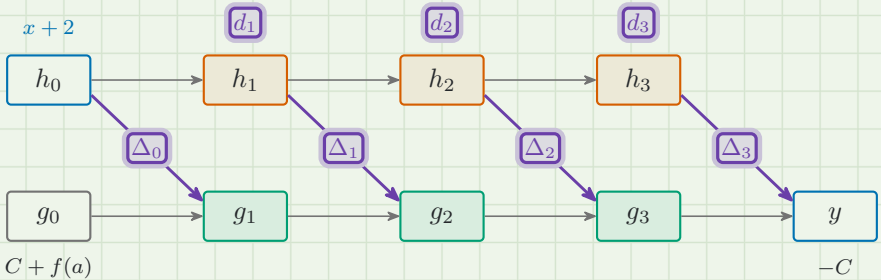
The knobs



So far. The network draws the zigzag exactly: the hinge lane makes the kinks, the sum lane adds the hinges up from a lift C, and the output pays the lift back.

Now. Which numbers set the network? Only the gaps d_ℓ and the slope changes Δ_ℓ: the knobs.

Two kinds of numbers set the whole network: the intercept of each hinge, its gap d_ℓ, and the slope change after each hinge's ReLU, Δ_ℓ, the weight on its arrow down to the sum lane. They are the *knobs*.

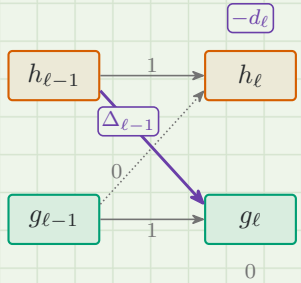


Gaps: one in each hinge neuron, $d_1, \ldots, d_k$. Each places a kink, one gap after the kink before.

Slope changes: one on each violet arrow, $\Delta_0, \ldots, \Delta_k$. Each bends the curve at its kink.

In all, k gaps and $k+1$ slope changes: $2k+1$ knobs, seven for the zigzag's three layers.

Written the usual way, with a weight matrix and a bias, layer ℓ has two free entries. A hinge neuron reads the hinge before it with weight 1 and the sum with weight 0; a running sum passes its total on with weight 1.

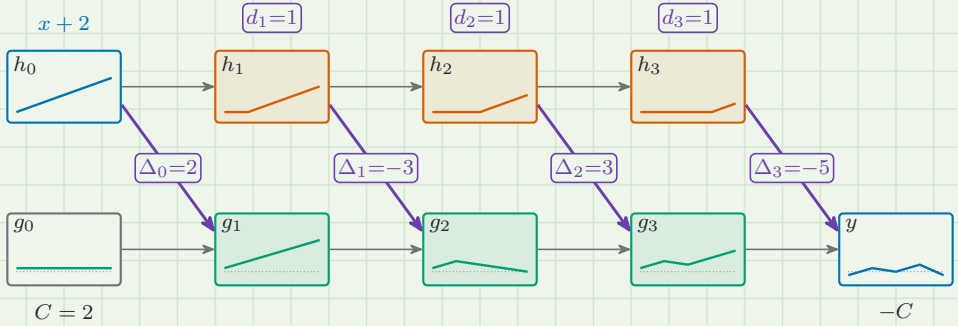


$$\begin{pmatrix} h_\ell \\ g_\ell \end{pmatrix} = \mathrm{ReLU}\left(\begin{pmatrix} 1 & 0 \\ \Delta_{\ell-1} & 1 \end{pmatrix} \begin{pmatrix} h_{\ell-1} \\ g_{\ell-1} \end{pmatrix} + \begin{pmatrix} -d_\ell \\ 0 \end{pmatrix} \right)$$

Violet: the two knobs of the layer. Grey: fixed, the same for every target and every layer.

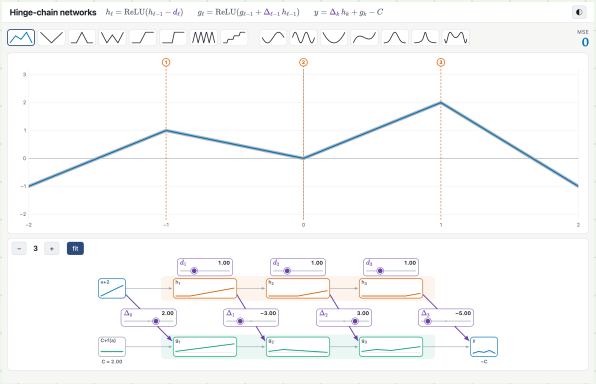
Two more numbers are no knobs: the start value $f(a)$, read off the target, and the lift C, the least that keeps every running sum above the floor. The page computes both.

Set the gaps to $d = (1, 1, 1)$ and the slope changes to $\Delta = (2, -3, 3, -5)$; then $f(a) = -1$ and $C = 2$ follow. Each box draws what its neuron outputs for x from -2 to 2.



Top: the hinges, with their kinks at -1, 0 and 1. Bottom: the running sums, each the zigzag so far, lifted by 2. Right: the output y, the zigzag.

The page *Hinge-chain networks* draws this network for fifteen targets: the plot on top, the network below, and every knob a violet slider where its number acts.



The zigzag in three layers, after fit: the error is 0.

Pick a target from the menu. It starts flat, at $f(a)$.

The buttons + and − add and remove a layer.

Move a slider, or drag a kink in the plot to set its gap.

fit sets every knob by the construction. A piecewise-linear target with k kinks is matched exactly by k layers.

A sine has no kinks to copy. With k layers, *fit* picks the k kinks whose straight pieces follow the sine most closely. The mean squared error falls from 0.42 with one layer to 2.3×10^{-4} with ten.



$k = 1$: error 0.42 $k = 2$: error 0.023 $k = 4$: error 0.0040

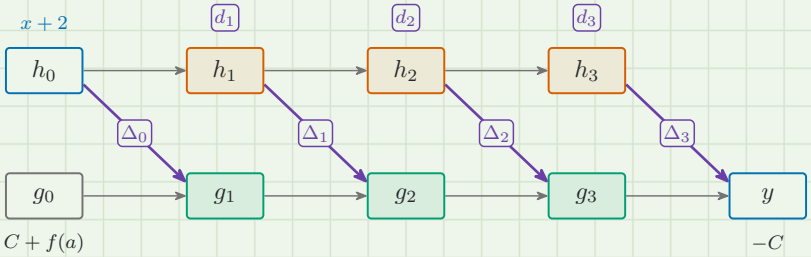
Grey: the sine. Blue: the network after fit. Dashed: its kinks.

Each layer adds one kink, so the network bends where the sine bends most. On the page, press + and then fit, and watch the error fall.

Entry 8

Summary

A hinge-chain network draws any continuous piecewise-linear function exactly. Its top lane makes the hinges, one kink per layer; its bottom lane adds them up.



$$h_\ell = \mathrm{ReLU}(h_{\ell-1} - d_\ell) \qquad g_\ell = \mathrm{ReLU}(g_{\ell-1} + \Delta_{\ell-1}\, h_{\ell-1}) \qquad y = \Delta_k\, h_k + g_k - C$$

The knobs are the gaps d_ℓ and the slope changes Δ_ℓ, $2k+1$ numbers. Every other weight is 1 or 0, and $f(a)$ and the lift C follow from the target.